LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 6

Course Code: INT396	Unit: 6
Course Title: UNSUPERVISED LEARNING	Unit Title: Evaluation and Applications of Unsupervised Learning
Course Outcome: CO6	Unit Mapped RBT: L3 (Apply)
Paper: Sample Subjective Questions	Total Questions: 10
Maximum Marks: 10 x 10 = 100	Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply internal validation to a clustering that has just been produced. (10 Marks)

(a) State two internal measures and describe what each of them computes.

(b) Use them to describe how you would compare a partition into three clusters with one into five.

[L3 - Apply]

Q2. Compare internal validation with external validation of a clustering. (10 Marks)

(a) State what each of the two requires before it can be computed.

(b) Use a case in which labels exist for a small subset of the records to describe how you would apply both.

[L3 - Apply]

Q3. Examine a clustering that scores well on internal measures but places almost every record into a single group. (10 Marks)

(a) State what the cluster sizes reveal that the score does not.

(b) Use this to describe what you would check and what you would change before running it again.

[L3 - Apply]

Q4. Apply a stability check to a customer segmentation. (10 Marks)

(a) Describe a resampling procedure that would test whether the segments are stable.

(b) Use the results of that procedure to state what would convince you the segments are genuine.

[L3 - Apply]

Q5. Demonstrate how a segmentation should be presented to the business team that will act on it. (10 Marks)

(a) State what you would show for each segment and why each item is needed.

(b) Use this to describe the limitations you would state alongside the result.

[L3 - Apply]

Q6. Use Principal Component Analysis and t-SNE together in a visualisation dashboard for a high-dimensional dataset. (10 Marks)

(a) State what each of the two contributes to the picture.

(b) Use this to describe the panels you would place on the dashboard and what each one would show.

[L3 - Apply]

Q7. Examine the ethical questions raised by segmenting customers on their behavioural data. (10 Marks)

(a) State two harms that could follow from acting on such a segmentation.

(b) Use these to describe two safeguards you would put in place before the system is used.

[L3 - Apply]

Q8. Apply the idea of interpretability to the result of an unsupervised method. (10 Marks)

(a) State why a clustering can be far harder to explain than a supervised prediction.

(b) Use cluster profiling to describe how you would make the segments understandable to a non-technical reader.

[L3 - Apply]

Q9. Relate the features chosen for a clustering to the segments it produces and to the fairness of those segments. (10 Marks)

(a) State how a proxy feature can reintroduce a protected attribute that was deliberately excluded.

(b) Use this to describe the checks you would carry out before the segmentation is deployed.

[L3 - Apply]

Q10. Compare a clustering built on the original features with one built on principal components. (10 Marks)

(a) State what changes about the result and about the distances the algorithm sees.

(b) Use this to state which of the two you would choose when the segments must be explained to a business team.

[L3 - Apply]

----- End of Sample Question Set -----